

Take-Home Challenge: Smart **Escalation** API

Role: Founding Full Stack Engineer **Company:** BolsterBiz **Expected Time:** 2-3.5 hours (please don't over-engineer) **Deadline:** 5 days from receipt

Context

We're building an AI-powered L1 customer support agent. The core challenge isn't getting the AI to answer questions — it's getting it to know when it can't answer reliably and should escalate to a human agent.

This challenge tests exactly that.

The Task

Build a single API endpoint and a minimal chat UI that does the following:

1. Accepts a customer question as input.
2. **Searches the** provided **help articles** (below) for relevant information.
3. Returns one of two responses:
 - An answer — grounded in the help articles, written in a friendly support-agent tone.
 - An escalation — a polite message telling the user a human agent will follow up.
4. **Includes a confidence indicator** in the API response (a score, label, or short explanation) showing why the system chose to answer or escalate.

That's it. No database, no auth, no ticketing system, no user accounts.

Technical Requirements

- Backend: Python (FastAPI preferred) or Node.js/TypeScript
- LLM: Use any LLM API you're comfortable with (Claude, OpenAI, etc.)
- Retrieval: Implement a basic RAG approach — embed the articles, retrieve relevant chunks, and pass them as context. Use any vector store or in-memory approach you prefer.
- Frontend: A simple chat interface in React. It doesn't need to be pretty — functional and clear is fine.
- Escalation logic: This is the most important part. We want to see your thinking behind when and why the agent escalates vs. answers.

Sample Help Articles

Use these five articles as the agent's knowledge base. They are for a fictional project management SaaS called **TaskFlow**.

Article 1: Getting Started with TaskFlow

TaskFlow is a project management tool designed for small and mid-size teams. After signing up, you'll land on your Dashboard, which shows all active projects. To create your first project, click the "+ New Project" button in the top-left corner, give it a name and description, and invite team members by email. Each project contains Boards, and each Board contains Tasks. You can drag and drop tasks between columns (To Do, In Progress, Done) to track progress. TaskFlow supports up to 50 team members on the Pro plan and 10 on the Free plan.

Article 2: Billing and Subscription Plans

TaskFlow offers three plans: Free, Pro (\$12/user/month), and Enterprise (custom pricing). The Free plan includes up to 3 projects, 10 team members, and 1 GB of file storage. The Pro plan includes unlimited projects, 50 team members, 50 GB storage, and priority email support. Enterprise includes everything in Pro plus SSO, audit logs, a dedicated account manager, and custom integrations. You can upgrade or downgrade at any time from Settings > Billing. When you upgrade mid-cycle, you'll be charged a prorated amount. Downgrades take effect at the next billing cycle. Refunds are available within 7 days of an upgrade if you haven't used any Pro-only features.

Article 3: Integrations

TaskFlow integrates with Slack, Google Calendar, GitHub, and Zapier. To connect an integration, go to Settings > Integrations and click "Connect" next to the app you want. Slack integration sends task notifications to a channel of your choice. Google Calendar integration syncs task due dates to your calendar. GitHub integration links pull requests to tasks — when a PR is merged, the linked task automatically moves to "Done." Zapier integration allows you to build custom automations with 3,000+ apps. Integrations are available on Pro and Enterprise plans only.

Article 4: Managing Team Permissions

TaskFlow has three permission roles: Owner, Admin, and Member. Owners have full control including billing and account deletion. Admins can create and manage projects, invite members, and configure integrations. Members can create and edit tasks within projects they're assigned to but cannot manage billing or integrations. To change someone's role, go to Settings > Team, click the three-dot menu next to their name, and select "Change Role." Only Owners can promote someone to Admin. There is only one Owner per workspace, but ownership can be transferred under Settings > Team > Transfer Ownership.

Article 5: Troubleshooting Common Issues

- **Tasks not syncing:** Try refreshing your browser or clearing your cache. If the issue persists, check your internet connection and ensure you're on the latest version of TaskFlow. If you're using the mobile app, force-close and reopen it.
 - **Can't invite team members:** Make sure you haven't hit your plan's member limit (Free: 10, Pro: 50). Also verify the email address is correct and hasn't already been invited.
 - **Integration not working:** Disconnect and reconnect the integration from Settings > Integrations. Make sure you've authorized TaskFlow in the third-party app's settings as well.
 - **Forgot password:** Click "Forgot Password" on the login screen. You'll receive a reset link within 5 minutes. Check your spam folder if it doesn't arrive. Reset links expire after 24 hours.
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What We're Looking For

Ranked by importance:

1. **Escalation quality** — Does the agent reliably answer when it should and escalate when it should? This is the single most important thing. We'll test with tricky edge cases (partial info, off-topic questions, ambiguous phrasing).
2. **Answer quality** — Are the answers accurate, grounded in the articles (not hallucinated), and written in a natural, helpful tone?
3. **Retrieval approach** — How do you chunk, embed, and retrieve context? We care more about your reasoning than the specific tools.
4. **Code clarity** — Is the code clean, readable, and structured in a way another engineer could pick up quickly?
5. **Prompt design** — How are you instructing the LLM? What guardrails have you built in?

We are NOT looking for:

- BeautifulUI design
 - Database setup or persistence
 - Authentication or user management
 - Over-engineered architecture
 - Comprehensive test suites (a few key tests are fine if you have time)
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Submission Instructions

1. Push your code to a private GitHub repo and invite [BolsterBiz-Hiring](#) as a collaborator.
 2. Deploy your working app to any free hosting platform (Render, Railway, Vercel, or similar) and include the live URL in your README. We'll use this to test your escalation logic directly — please make sure it stays live for at least 7 days after submission. Deployment to a *free tier* typically takes *10-15 minutes* and *is included* in the time *estimate*.)
 3. Include a **README** with:
 - The live deployed URL (this is how we'll primarily evaluate your submission)
 - Setup instructions for running locally as a backup (should take less than 5 minutes)
 - A brief explanation (3-5 paragraphs) of your escalation logic — why does the agent escalate when it does?
 - Any trade-offs you made given the time constraint
 - What you'd improve with more time
 4. Optional but appreciated: A 2-minute Loom video walking through your solution. This is not required but helps us understand your thinking.
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A Note on Time

We respect your time. This is scoped to be completable in **2-3.5 hours** by someone with the right experience. The deployment step is intentionally lightweight — connecting a GitHub repo to a free hosting platform should take no more than 15 minutes.

If you find yourself going past 4 hours, stop — submit what you have with a note about what's remaining. We'd rather see a well-thought-out partial solution than a rushed complete one.

Questions?

If you have any questions about the task, reply to this email. Good luck — we're genuinely excited to see how you approach this.

— The BolsterBiz Team